

CSIS 4495 – Applied Research Project

Progress Report 3

GREENPATH – Sustainable Products Website with AI Chatbot Integration

FRANCIS AWOYALE (300356743)

Work Date/Hours Log

The table below shows summary of the work done till date with the respective time spent on each task.

Date	Number of Hours	Description of Work Done
2025-10-09	3	Reviewed RootLayout.tsx structure and verified sticky header functionality across pages.
2025-10-10	3	Finalized responsive navigation pills and added divider border for visual separation.
2025-10-11	3	Built country-ranking table on Analytics page using dynamic dataset view.
2025-10-12	2	Removed duplicate Navbar component and kept RootLayout.tsx as the sole active layout.
2025-10-13	3	Resolved Tailwind and VS Code warnings through workspace configuration.
2025-10-14	3	Developed the initial Chatbot UI structure with sample conversation mock-ups.
2025-10-15	3	Integrated Chatbot backend with AWS Lambda and OpenAI API for live responses; encountered CORS and response-latency issues.
2025-10-16	2	Integrated Chatbot backend with AWS Lambda and OpenAI API for live responses; encountered CORS and response-latency issues. (continuation)
2025-10-17	3	Fixed CORS errors by adding response headers in Lambda and used asynchronous promises to reduce latency.
2025-10-18	3	Added typing indicator and scroll persistence to Chatbot UI; resolved duplicate message rendering bug.
2025-10-19	3	Performed frontend Chatbot tests and enhanced responsiveness on mobile devices.
2025-10-20	3	Implemented fallback messages and user error handling for API timeouts.
2025-10-21	2	Integrated Chatbot launch button into Home, Categories and Analytics pages.
2025-10-22	2	Created Analytics Machine Learning (AnalyticsML) sub-page as a placeholder for future model integration.

Description of Work Done

These past weeks, I achieved major progress in both the AI Chatbot integration and the Analytics module expansion of the GreenPath application. The Chatbot was implemented using React for the frontend and an AWS Lambda function integrated with the OpenAI API for backend processing. Several issues were encountered during development, including CORS errors, delayed API responses, and duplicate message rendering. These were resolved by configuring appropriate CORS headers in the Lambda response, restructuring asynchronous API calls to improve response speed, and optimizing state logic to prevent duplicate outputs. Enhancements such as typing indicators, scroll retention, and mobile responsiveness were also added to improve usability. In parallel, the Analytics page was extended with a dynamic country-ranking table that displays emission data by sector and year, and a new Analytics Machine Learning sub-page was created to serve as a foundation for the next milestone involving predictive data analysis. General UI polishing continued throughout the period, including standardization of the sticky header, navigation buttons, and Tailwind styling to maintain design consistency across all pages. Overall, this phase advanced GreenPath into a functional, data-driven, and AI-enhanced platform ready for subsequent machine-learning implementation.

Repo Check in of Implementation Completed

The files/folders I have checked in the repo since report 2 are as follows:

Recent committed files under Implementation/greenpath/ include:

src/pages/Chat.tsx – React Chatbot page managing conversation UI and API calls.

backend/lambda/Chat.js – AWS Lambda function integrated with OpenAI API and CORS headers.

src/pages/Analytics.tsx – New country-ranking table view by sector and year.

src/pages/AnalyticsML.tsx – Machine Learning sub-page created for future model integration.

src/layout/RootLayout.tsx – Finalized global layout with sticky header and navigation design.

src/index.css – Updated Tailwind directives and consistent white link color.

.vscode/settings.json – Workspace configuration enhancements for Tailwind IntelliSense.

public/images/categories/ – Added more image assets for hero and gallery sections.